

CHUMPHON, Thailand

WMO: 485170

Lat: 10.4990N

Lon: 99.1884E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	67.7	69.8	61.7	82.3	73.2	64.2	89.9	74.8	18.1	80.4	16.5	81.1	0.3	300	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	15.7	94.9	80.0	93.5	79.9	92.4	79.6	82.4	90.3	81.6	89.6	81.0	89.0	5.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	159.2	86.5	79.2	151.8	85.9	78.9	149.9	85.7	46.4	90.1	45.4	90.3	44.7	89.4	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.8	11.2	63.6	97.7	3.2	1.9	61.3	99.1	59.4	100.2	57.6	101.3	55.3	102.7		
(5)			61.4	84.2	2.9	1.2	59.3	85.1	57.6	85.8	56.0	86.5	53.9	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	79.2	80.7	82.9	84.8	84.2	83.2	82.3	82.4	82.0	81.4	80.4	78.9
	DBStd	2.79	2.24	2.02	2.07	2.11	2.35	2.22	2.19	2.13	1.91	2.01	2.12	2.56
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11623	904	860	1019	1043	1059	996	1000	1003	959	974	911	895
	CDD65	6148	439	440	554	593	594	546	535	538	509	509	461	430
	CDH74	62037	3874	4453	6197	7076	6586	5800	5332	5376	4875	4712	4144	3612
(13)	CDH80	22732	1227	1700	2709	3308	2693	2149	1802	1791	1555	1489	1258	1051
(14)	Wind	WSAvg	3.2	3.1	2.8	2.8	2.6	3.1	3.6	3.9	3.4	2.6	3.0	3.7
(15) Precipitation	PrecAvg	91.8	2.6	1.6	4.3	3.9	9.2	10.0	11.2	11.7	10.7	11.6	9.8	5.4
	PrecMax	114.1	12.2	8.0	20.1	9.1	17.8	17.6	18.9	29.6	17.4	24.2	21.0	19.4
	PrecMin	68.5	0.2	0.0	0.0	0.0	4.1	3.3	4.1	4.7	5.5	3.4	2.2	0.0
	PrecStd	11.8	2.4	1.9	5.4	2.7	3.5	3.3	3.8	5.2	3.0	4.6	5.0	5.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.4	93.2	95.6	97.2	96.2	94.5	93.2	92.7	92.6	91.7	91.2	90.2
		MCWB	76.9	77.9	78.3	80.3	80.6	80.0	79.0	78.7	79.5	79.4	79.0	77.8
	2%	DB	88.7	90.9	93.4	95.2	94.1	92.6	91.5	91.2	90.8	90.1	89.3	88.4
		MCWB	77.5	78.0	79.2	80.3	80.6	79.5	78.7	78.7	79.0	79.3	78.8	77.8
	5%	DB	87.5	89.5	91.9	93.7	92.5	91.1	89.9	89.8	89.3	88.9	87.9	87.0
		MCWB	77.2	77.8	79.1	80.3	80.4	79.1	78.7	78.6	78.8	79.1	78.8	77.3
	10%	DB	86.1	88.0	90.2	92.2	90.7	89.3	88.1	88.1	87.6	87.4	86.4	85.6
		MCWB	76.9	77.6	78.8	80.2	80.1	78.9	78.6	78.5	78.5	78.9	78.2	76.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.7	81.8	83.3	83.7	82.7	81.9	81.7	81.6	81.7	81.5	80.8
		MCDB	85.9	88.9	90.1	92.8	91.2	89.3	87.6	88.1	87.9	87.3	86.2	86.6
	2%	WB	79.2	79.6	80.8	82.1	82.4	81.1	80.5	80.2	80.3	80.7	80.4	79.6
		MCDB	85.7	87.8	89.7	91.8	90.8	88.8	87.8	87.6	87.7	87.7	86.6	85.7
	5%	WB	78.4	78.9	80.0	81.4	81.3	80.1	79.6	79.4	79.5	79.9	79.6	78.7
		MCDB	85.2	87.0	88.9	90.9	89.7	87.9	86.8	86.7	86.6	86.6	85.7	84.8
	10%	WB	77.7	78.2	79.5	80.7	80.5	79.4	78.9	78.8	78.8	79.1	78.9	77.8
		MCDB	84.2	86.0	88.2	90.1	88.6	86.6	86.0	86.0	85.7	85.6	84.9	83.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	15.6	15.9	15.7	13.5	12.5	11.9	11.7	11.8	11.8	11.8	13.5
		MCDBR	16.3	17.5	18.1	17.7	16.1	15.0	14.6	14.2	14.2	14.4	14.5	15.4
		MCWBR	6.9	6.7	6.6	6.1	5.5	4.9	4.8	4.6	4.8	5.5	6.0	6.5
	5% WB	MCDBR	14.5	15.6	16.3	16.5	15.0	13.6	13.0	12.8	12.9	13.2	12.9	13.3
		MCWBR	6.6	6.6	6.7	6.4	6.0	5.4	5.3	5.1	5.3	5.8	6.0	6.2
(40) Clear-Sky Solar Irradiance	taub		0.495	0.515	0.520	0.483	0.454	0.451	0.450	0.444	0.430	0.462	0.484	0.483
	taud		2.152	2.096	2.124	2.243	2.370	2.410	2.396	2.405	2.437	2.322	2.230	2.206
	Ebn at Noon		258	258	258	265	268	265	266	272	279	269	259	259
	Edh at Noon		48	53	52	46	39	37	38	38	37	41	44	45
(44) All-Sky Solar Radiation	RadAvg		1525	1739	1821	1845	1582	1474	1403	1447	1474	1381	1353	1387
	RadStd		117	102	140	128	138	80	128	144	111	134	136	119

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.41	N/A	N/A	N/A	+0.49	+0.56	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.34	+0.77	N/A	+0.67	+0.54	+0.54	N/A	N/A	+137	+137

Nomenclature: See separate page